

Astrid Greene

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EDUCATION

University of Michigan

B.S. in Computer Science, Minor in French

Ann Arbor, MI

Expected Graduation: May 2028

- GPA: 3.7/4.0, University Honors
- Coursework: Data Structures and Algorithms, Machine Learning, Computer Organization, Object Oriented Programming, Discrete Mathematics, Linear Algebra, Calculus I-II, Computer Pragmatics, Programming Concepts
- Activities: Girls in Electrical Engineering and Computer Science, Tech+ Consulting and Development, EECS 201 IA

EXPERIENCE

Coretek

Software Engineering Intern

May 2026 – Present

Farmington Hills, MI

- Built Python Azure Functions for automated expense validation, implementing a modular rule engine across 5 files with 19 policy rules, worst-wins result aggregation, and configurable thresholds deployed on a Consumption plan
- Wrote 200+ test cases across two Python test suites using an independent reference oracle pattern, covering boundary values, cross-line rules, and field-robustness edge cases against local and deployed function endpoints
- Designed a validation architecture integrating Salesforce triggers, OCR, and Python Azure Functions, processing 5,000+ annual expense lines with confidence-scored OCR fallback and a 3-stage approval routing system

Tech Plus Development

Feb 2026 – May 2026

Software Engineer

Ann Arbor, MI

- Designed and implemented a role-based authentication system using Supabase Auth and PostgreSQL, writing 10+ Row Level Security policies to enforce granular access control across 3 user roles (admin, member, recruit)
- Developed a full-stack internal platform in React, TypeScript, and Vite, featuring a member management, attendance tracking, and event scheduling features, serving 50+ active club members across 6 project teams
- Configured full-stack deployment pipeline using GitHub, Supabase, and Vercel, managing environment variables and API keys across both development and production environments to support continuous deployment workflows

Tech Plus Consulting

Jan 2026 – May 2026

Technical Analyst

Ann Arbor, MI

- Built an AI policy chatbot by migrating 200+ client policy documents into a structured Google Drive and connecting U-M Maizey's REST API to embed a searchable chatbot widget directly on the client's password-protected site
- Resolved data ingestion issues caused by inconsistent document formatting, access-restricted pages, and duplicate file versions, cleaning and standardizing source data across 15+ file types to improve retrieval accuracy and reliability
- Conducted 6+ stakeholder meetings over a 4-week sprint to define system requirements, identify access constraints, and scope chatbot functionality, reducing the initial feature set by 40% to prioritize high-value policy retrieval

Morgan State University

June 2023 – Aug 2023

Research Assistant

- Co-authored paper "Debunking The Curse of Dimensionality in a K-Nearest Neighbors Classification Problem" with advisor Dr. Eric Sakk, selected as a national Semi-Finalist in the Junior Science and Humanities Symposium
- Researched "Curse of Dimensionality" in k-Nearest Neighbors, running controlled Python experiments to show that in uniform, hard-confidence data sets, increasing dimensionality can improve k-NN classification performance
- Designed k-NN experiments in Python on datasets of 1000+ points, varying k-values and dimensions (2D–15D) to test classification accuracy, with NumPy, Scikit-learn, and Matplotlib for data generation, training, and visualization

PROJECTS

Order Book Simulator

Oct 2025

- Built a central limit order book matching engine in C++ using heap-backed bid/ask books for low-latency trade execution, processing 1M+ orders and executing 760K+ trades in < 10 seconds with O(log n) matching complexity
- Implemented limit and market order types with real-time bid-ask spread tracking, partial fill logic, and order cancellation, validating engine accuracy against 500K+ expected trade outputs with automated test scripts

Deep Learning Dog Breed Classification

Mar 2026

- Developed custom PyTorch CNN and Vision Transformer architectures for 10-class image classification across a 9000-image dataset, implementing transformer encoders, multi-head self-attention, and scaled dot-product attention
- Built transfer learning pipelines with Adam optimization, checkpoint serialization, tensor normalization, and 256-patch embeddings, training models over 10,000+ update iterations to improve convergence and performance

ICU Mortality Prediction Model

Feb 2026

- Engineered a clinical ML pipeline over 12,000+ ICU admissions and 40+ physiological variables, transforming sparse EHR time-series data through normalization, imputation, statistical aggregation, and feature engineering workflows
- Executed 5-fold cross-validation and 1,000+ bootstrap resampling iterations across logistic regression, kernel ridge regression, and RBF kernel models, benchmarking AUROC, sensitivity, and specificity under imbalance conditions

TECHNICAL SKILLS

Languages: C/C++, C#, Java, Python, JavaScript/TypeScript, HTML/CSS, SQL, R, Bash, YAML, LaTeX

Developer Tools: Git, Linux, Node.js, REST APIs, PostgreSQL, Azure, Vercel, Azure Functions, Power Automate

Interests: Avid Sudoku Solver, Card Game Enthusiast, Dedicated Tennis Player, Music Lover & Passionate Musician